

Project Initialization and Planning Phase

Date	27 June 2024
Team ID	XXXXXX
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To build a reliable, accessible web application that predicts flood risk in advance and delivers timely alerts so at-risk communities can prepare and respond before flooding occurs.
Scope	The project covers secure user registration and login, an ML-based flood risk prediction feature, an AI chatbot for flood-safety guidance, and email-based verification and alerts, all deployed as a single Flask web application.
Problem Statement	
Description	Residents of flood-prone regions often lack an accessible, real-time tool to gauge their flood risk, relying instead on generic, delayed government bulletins that are not localized to their situation.
Impact	Without early, localized warning, residents have little time to prepare, increasing risk to life and property. A predictive, easy-to-use tool can give people the lead time needed to move valuables, evacuate, or seek help.

Proposed Solution	
Approach	Use a machine learning model trained on rainfall and river-level data to estimate flood risk, and serve predictions through a secure, easy-to-use Flask web application accessible from any device.
Key Features	- ML-based flood risk prediction using rainfall and river-level data. - Secure authentication via Google OAuth and manual signup with email/phone verification. - Gemini-powered chatbot widget offering flood-safety guidance and app help. - Automated email alerts and verification via Brevo integration.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard CPU instance (no GPU required; model is lightweight)
Memory	RAM specifications	2 GB (Render free/standard web service tier)
Storage	Disk space for data, models, and logs	1 GB SSD
Software		
Frameworks	Python frameworks	Flask

Libraries	Additional libraries	scikit-learn, pandas, numpy, Flask-Login, Flask-Mail / Brevo SDK, google-auth
Development Environment	IDE	VS Code
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv